

Learning Fault-tolerant Quadrupedal Locomotion in Rough Terrains

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Abstract—Recently, quadrupedal robots are widely deployed in inspection, rescue, and other field operations where human access is difficult or dangerous. In these missions, even a single joint failure can severely restrict the robot’s mobility and lead to mission failure, particularly on uneven and unstructured terrain. Robust fault-tolerant locomotion is therefore critical for reliable operation in such environments. However, existing methods typically consider only a limited set of joint failures or terrain conditions, making it difficult to apply a single policy across diverse failure–terrain combinations. To address this limitation, we present a unified framework for blind quadrupedal locomotion across various terrains under normal, single-joint, and selected diagonal two-joint failure conditions. The framework consists of three main components: 1) joint-failure and terrain condition generation, 2) Reward-Aware Population Allocation (RPA), and 3) failure-aware policy conditioning. The first component creates diverse training scenarios by combining various terrains and joint failures, together with a folded-leg posture to reduce unwanted terrain contact from affected legs. RPA then dynamically reallocates the failure and terrain populations independently according to category returns, enabling more focused training on difficult conditions. Finally, the framework infers body velocity and joint-failure states from proprioceptive histories and conditions a unified policy on these estimates, enabling failure-specific locomotion behaviors. Extensive simulation and real-robot experiments demonstrate that our framework enables a single policy to maintain locomotion on deformable, uneven, moving, and stair terrains under single-joint and selected diagonal two-joint failure conditions without additional real-robot training.

I. INTRODUCTION

Quadrupedal robots have demonstrated reliable locomotion over challenging and unstructured terrain [1], [2]. However, a joint failure can severely degrade locomotion performance, especially on discontinuous terrain such as stairs [3], [4]. Reduced actuation can lead to poor foot placement, unwanted terrain contact, or a fall. Therefore, practical fault-tolerant locomotion should consider both joint failures and the terrain on which they occur.

Existing studies address joint failures using control strategies designed for individual failure conditions [5]–[7], fault detection followed by control adaptation [8], or policies trained with actuator failures [9]–[12]. Control strategies designed for individual failure conditions require a suitable controller for each expected case. Switching based on fault diagnosis also requires a higher-level module to identify the current condition and select the appropriate controller. Its performance therefore depends on reliable fault identification and may be affected by diagnosis errors or switching delays.

A single policy avoids the need to maintain a separate controller for each failure condition. However, learning many failure and terrain conditions within one policy introduces a different problem. A limited number of parallel environments

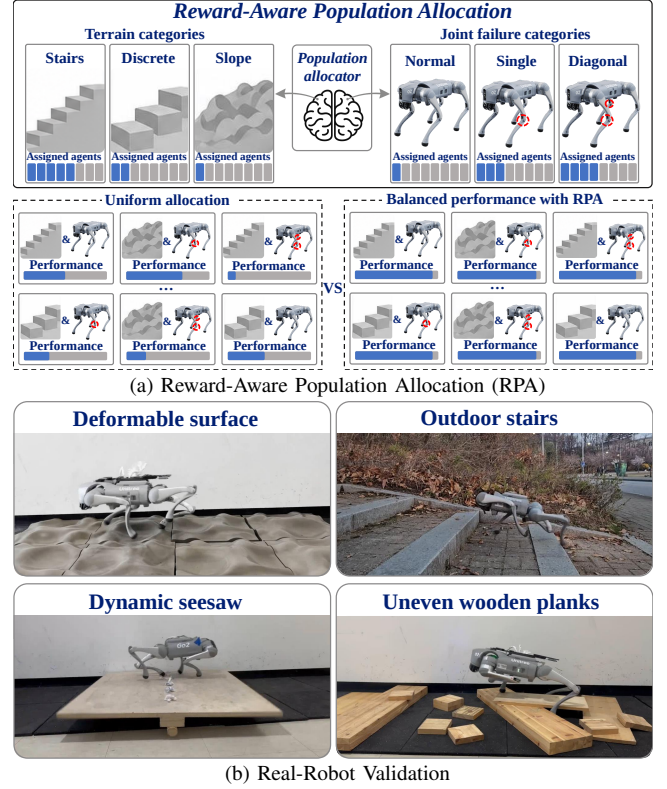


Fig. 1: Overview of the proposed method and real-robot validation. (a) Reward-Aware Population Allocation (RPA) assigns more parallel environments to failure and terrain categories with lower returns, reducing performance differences across training conditions. (b) The unified policy maintains locomotion under single-joint and selected diagonal two-joint failures on deformable, uneven, dynamic, and stair terrains.

and a fixed training budget must be shared across all conditions. As the number of conditions increases, each condition receives less training experience. Moreover, these conditions can have different learning difficulties. Difficult conditions may therefore remain undertrained when all conditions receive the same training resources.

A unified policy must also identify the current failure state and adapt its behavior accordingly. Changes in robot motion can result from terrain contact, external disturbances, or joint torque loss. Therefore, a single observation may not be sufficient to distinguish these causes. Previous studies have used temporal observations to estimate joint or actuator states [8], [10], [13]. Recent proprioceptive history can therefore provide useful information about persistent changes caused by a joint failure.

To address these problems, we propose the unified framework shown in Fig. 1. We construct the training domain using normal operation, single-joint failures, and multiple

terrain conditions. We further include selected diagonal two-joint failures to extend the policy beyond single-joint failures. Trotting locomotion relies on diagonal support pairs, making simultaneous joint failures on opposing diagonal legs particularly challenging. Random Joint Masking [12] is used to model joint torque loss. A predefined folded-leg posture reduces unwanted contact between the affected leg and rough terrain.

We then introduce Reward-Aware Population Allocation (RPA), which adjusts the number of parallel environments according to the current performance of failure and terrain categories. Categories with lower returns receive more training experience. This provides more training opportunities for conditions that remain difficult during training.

Finally, we use history-based state estimation to condition a single locomotion policy. A history encoder processes recent observation histories, from which a body-velocity estimator and a fault-state classifier estimate body velocity and joint failure states. These estimates are provided to the actor so that the policy can adapt its behavior to different failure conditions without explicit policy switching.

The learned policy is transferred to a Unitree Go2 robot without additional real-robot training. As shown in Fig. 1(b), we evaluate the policy on a deformable surface, uneven wooden planks, a dynamic seesaw, and outdoor stairs under single-joint and selected diagonal two-joint failures.

The main contributions of this work are summarized as follows:

- We present a unified framework for blind quadrupedal locomotion under normal, single-joint, and selected diagonal two-joint failures across diverse terrains using a single policy.
- We introduce Reward-Aware Population Allocation, which adjusts the training population according to the performance of failure and terrain categories. This provides more training experience for difficult conditions and reduces performance variation across categories.
- We introduce a history-based state estimator that infers body velocity and joint failure states from recent observation histories and conditions the locomotion policy. This allows a single policy to adapt to different failures without explicit policy switching.
- Simulation and real-robot experiments show that the proposed framework maintains locomotion on deformable, uneven, dynamic, and stair terrains under single-joint and selected diagonal two-joint failure conditions.

II. RELATED WORK

A. Model-Based Fault-Tolerant Locomotion

Model-based fault-tolerant control modifies the robot model, contact constraints, or gait after a failure. Early studies investigated fault-tolerant gaits under locked-joint failures [14], [15]. Later work used posture optimization and whole-body control to maintain quadrupedal locomotion under joint locking [5]. Other studies designed fault-tolerant gaits for quadrupedal robots with one or two locked legs [6], [7].

Recent work has combined fault detection with model-based control. Liu *et al.* use a GRU-based detector to identify partial joint failures and estimate joint torque retention rates [8]. The estimated fault information is then incorporated into an MPC and whole-body control framework for fault compensation. However, these approaches generally require the failure state to be identified and reflected in the control model or constraints. They may also require repeated online optimization. Learning-based methods provide an alternative by learning recovery behaviors directly from failure experience.

B. Learning-Based Fault-Tolerant Locomotion

Learning-based methods expose a policy to actuator failures during training so that recovery behaviors can be learned without explicitly modifying the robot model. Earlier studies used reinforcement learning to recover locomotion under joint or motor failures [3], [9]. Meta-reinforcement learning has also been used to adapt quadrupedal locomotion to motor-stuck conditions [16]. Random Joint Masking trains a single policy for normal and impaired locomotion by introducing joint failures during training and using observation histories to infer joint states [12].

Recent studies have expanded fault-tolerant learning to more diverse terrain and failure conditions. DreamFLEX estimates joint faults and uses the estimated state to adapt locomotion on rough terrain [10]. FT-CPG combines reinforcement learning with central pattern generators to handle concurrent joint locking and power loss within a leg [11]. Matsuura and Sato train a single model to handle sudden free-swinging failure of any one of the twelve joints [17]. Contrastive forward prediction has also been used to adapt locomotion across different terrains and joint damage conditions, including unseen locked-joint failures [18]. Recent work on adaptive gait timing uses proprioceptive observations and a learned gait frequency to adapt to actuator power loss and terrain variation [19].

Multiple-joint failures have also received increasing attention. AcL distills fault-specific teacher policies into a single student policy and considers multiple faulty joints across one or two legs [20]. Fault-tolerant learning has also been studied in humanoid locomotion. TOLEBI classifies joint conditions online and learns bipedal locomotion under joint locking and power loss [13].

These studies have improved fault estimation and recovery behavior while broadening the range of failure conditions that can be handled by a learned policy. Our work focuses on a different problem that arises when many failure and terrain conditions are learned together. These conditions can have different learning difficulties while sharing a fixed number of parallel environments. We therefore introduce Reward-Aware Population Allocation, which assigns more environments to failure and terrain categories with lower returns. The policy is also conditioned on failure information inferred from proprioceptive histories, allowing a single actor to adapt to different failure conditions without explicit policy switching.

III. METHOD

A. Problem Formulation

We consider a quadrupedal locomotion problem in which a single policy tracks velocity commands under different joint failure and terrain conditions. Let $\mathcal{F}$ denote the set of failure configurations, including normal operation, single-joint failures, and selected diagonal two-joint failures. Let $\mathcal{T}$ denote the set of terrain configurations. A training condition is defined as $c = (\phi, s)$, where $\phi \in \mathcal{F}$ is a failure configuration and $s \in \mathcal{T}$ is a terrain configuration. The policy is trained across conditions constructed from these two sets.

At time step t , the actor receives the current policy observation o_t , an estimated body velocity $\hat{v}_t$, and a predicted fault-state vector $\hat{y}_t$. The latter two are inferred from recent observation history. The observation o_t contains the commanded planar velocity, commanded yaw velocity, body angular velocity, projected gravity, joint positions, joint velocities, and the previous action. The sequence of the H most recent observations is denoted by $o_t^H = (o_{t-H+1}, \dots, o_t)$, where H is the history length used by the history encoder.

The policy is denoted by π_θ , where θ represents its parameters. The policy produces an action $a_t \in \mathbb{R}^{12}$. The action is scaled and added to the default joint configuration to obtain the reference joint position,

$$q_t^{\text{ref}} = q^{\text{default}} + s_a a_t, \quad (1)$$

where q^{default} denotes the default joint configuration and s_a is the action scaling factor. The reference joint position is then converted to joint torque using a PD controller,

$$\tau_t^{\text{PD}} = K_p (q_t^{\text{ref}} - q_t) - K_d \dot{q}_t, \quad (2)$$

where τ_t^{PD} is the torque vector generated by the PD controller. The matrices K_p and K_d denote the proportional and damping gains, respectively. The vectors q_t and $\dot{q}_t$ denote the measured joint position and joint velocity.

The policy and value function are trained using PPO [21]. The objective is to maximize the expected discounted return,

$$J(\pi_\theta) = \mathbb{E}_{\xi \sim p(\xi|\pi_\theta)} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right], \quad (3)$$

where $J(\pi_\theta)$ is the expected return of the policy. The symbol ξ denotes a trajectory generated by the policy, and $p(\xi|\pi_\theta)$ denotes its trajectory distribution. The value T denotes the episode length, r_t is the reward at time step t , and γ is the discount factor.

The reward function encourages command tracking and stable locomotion. The complete reward terms and their weights are given in Table I. The reward consists of linear velocity tracking, angular velocity tracking, upright posture maintenance, torque minimization, foot slip reduction, action smoothing, and foot clearance.

In Table I, v_{xy} and v_{xy}^{cmd} denote the measured and commanded planar velocities. The values ω_z and ω_z^{cmd} denote the measured and commanded yaw velocities. The vector

TABLE I: Reward terms and weights used for policy training. The shaded foot-clearance term is applied only during normal operation.

Reward term	Equation	Weight
Linear velocity tracking	$\exp(-\ v_{xy} - v_{xy}^{\text{cmd}}\ ^2 / 0.25)$	1.5
Angular velocity tracking	$\exp(-(\omega_z - \omega_z^{\text{cmd}})^2 / 0.25)$	1.0
Upright posture maintenance	$\exp(-\ \theta_{\text{roll,pitch}}\ ^2 / 0.25)$	0.5
Torque minimization	$\ \tau_t\ ^2$	-0.0001
Foot slip reduction	$\ v_{xy}^{\text{foot,contact}}\ ^2$	-0.05
Action smoothing	$\ a_t - a_{t-1}\ ^2$	-0.05
Foot clearance (normal only)	$(p_{f,z,k}^{\text{des}} - p_{f,z,k})^2 \cdot \ v_{f,xy,k}\ $	-0.5

$\theta_{\text{roll,pitch}}$ contains the body roll and pitch angles. The vector τ_t contains the applied joint torques. The vector $v_{xy}^{\text{foot,contact}}$ denotes the planar foot velocity during contact. The vectors a_t and a_{t-1} are the current and previous actions. The values $p_{f,z,k}^{\text{des}}$ and $p_{f,z,k}$ denote the desired and measured vertical positions of foot f at foot-step phase k , and $v_{f,xy,k}$ denotes its planar velocity. The symbols f and k denote the foot index and foot-step phase index, respectively.

We intentionally minimize condition-specific reward shaping so that the policy can adapt its motion across different failure and terrain conditions without being strongly constrained to predefined recovery patterns. The foot-clearance reward is applied only during normal operation. Since the same policy is trained across normal and failure conditions, it is still exposed to foot-clearance supervision during normal locomotion. However, directly enforcing a fixed foot-clearance objective under failure conditions can overly constrain recovery motions, particularly on discontinuous terrain such as stairs, where the policy may need to deviate from the nominal foot-clearance pattern. Therefore, the foot-clearance term is disabled under failure conditions.

B. Framework Overview

The proposed framework trains a single locomotion policy across the failure and terrain conditions defined above. Fig. 2 provides an overview of the overall training framework, including condition generation, adaptive population allocation, fault-state estimation, and policy optimization.

First, the training domain is constructed from normal operation, all single-joint torque-loss conditions, selected diagonal two-joint failures, and multiple terrain configurations. At each environment reset, a failure configuration and a terrain configuration are assigned to each parallel environment.

Second, Reward-Aware Population Allocation (RPA) adjusts the training population according to the current performance of failure and terrain categories. Categories with lower returns receive more parallel environments, providing additional training experience for conditions that remain difficult during training.

The policy is further conditioned on information inferred from recent observation histories. A history encoder maps the observation history to a latent representation, which is used by a body-velocity estimator and a fault-state classifier. The estimated body velocity and predicted fault states are provided to the actor together with the current policy

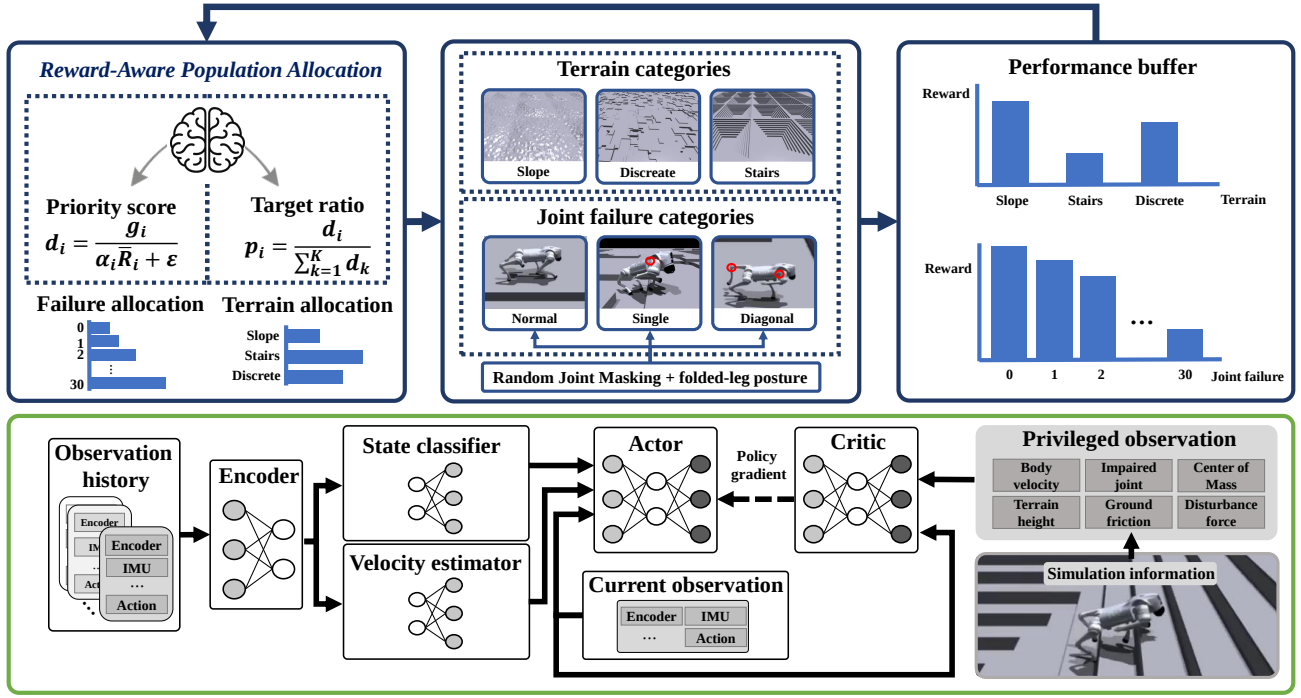


Fig. 2: Overview of the proposed training framework. Reward-Aware Population Allocation (RPA) independently adjusts the failure and terrain populations according to their category returns. A history encoder maps recent observation histories to a latent representation, from which a body-velocity estimator and a fault-state classifier estimate body motion and joint failure states. These estimates are provided to the actor together with the current policy observation. Policy optimization uses an asymmetric actor-critic structure, where privileged simulation information is provided only to the critic during training.

observation. This allows the same actor to adapt its locomotion behavior to different failure conditions without explicit policy switching.

Policy optimization uses an asymmetric actor-critic structure. The actor receives only information available during real-robot deployment, whereas the critic additionally receives privileged simulation information. The privileged information is used only during training for value estimation and is not required during deployment.

C. Joint Failure and Terrain Condition Generation

The training domain is constructed from a set of failure configurations and a set of terrain configurations. The goal is to expose the unified policy to diverse failure and terrain conditions while keeping the training domain manageable.

Failure configurations. The failure set $\mathcal{F}$ contains normal operation, all single-joint torque-loss conditions, and selected diagonal two-joint failures. Each single-joint condition disables one of the actuated joints. The diagonal two-joint conditions place failures on opposing diagonal legs, directly affecting the diagonal support structure used during trotting locomotion.

Including all possible two-joint failures would substantially increase the number of training configurations and reduce the training experience available for each condition under a fixed training budget. We therefore include selected diagonal two-joint failures, which provide challenging recovery conditions due to the diagonal support structure of trotting locomotion.

Terrain configurations. The terrain set $\mathcal{T}$ contains Flat, Discrete, Smooth Slope, Rough Slope, Stairs Up, and Stairs Down. The selected terrains cover both continuous and discontinuous contact conditions. Sloped terrain mainly changes body orientation and load distribution, whereas discrete terrain and stairs additionally require adaptation in foot placement and clearance. These discontinuous terrains become particularly challenging under joint failures because the impaired leg can move passively and make unintended contact with terrain edges. We therefore include stair ascent and descent as important training conditions for learning recovery behaviors under joint failures.

At each environment reset, one failure configuration and one terrain configuration are assigned to each environment. The failure and terrain populations are adjusted independently by RPA, as described in Sec. III.D.

Failure realization. We use Random Joint Masking [12] to realize joint torque loss during training. Random Joint Masking exposes the policy to different joint failure conditions by disabling the torque of selected joints. However, direct torque masking leaves the failed joint passive, which can cause the affected leg to hang below the body. This becomes problematic on discontinuous terrain such as stairs, where the impaired leg can strike terrain edges and disturb locomotion.

As illustrated in Fig. 3, we therefore introduce a folded-leg posture to reduce interference between the affected leg and the terrain. Rather than constraining the recovery motion of the entire robot, the actions of the joints in the affected

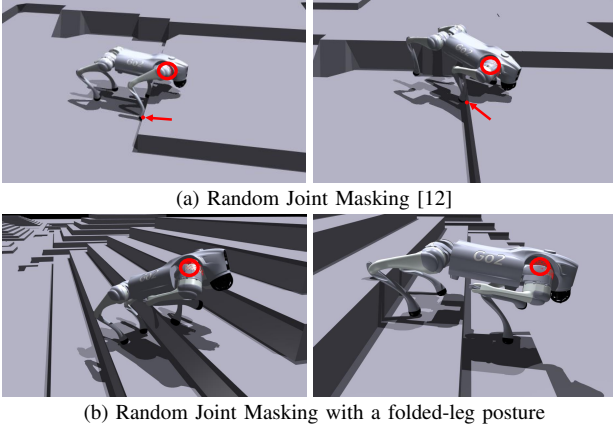


Fig. 3: Effect of the folded-leg posture during stair traversal. (a) Under conventional Random Joint Masking, the affected leg can hang below the body and contact a stair edge. (b) The folded-leg posture keeps the affected leg closer to the body, reducing unintended contact with the terrain.

leg are replaced with predefined folded actions.¹ For joint j in an affected leg,

$$\tilde{a}_t(j) = \begin{cases} a_{\text{folded}}(j), & \text{if } j \text{ belongs to an affected leg,} \\ a_t(j), & \text{otherwise,} \end{cases} \quad (4)$$

where $a_t(j)$ is the action produced by the policy, $a_{\text{folded}}(j)$ is the predefined folded action, and $\tilde{a}_t(j)$ is the action after replacement. For failure conditions, a_t in (1) is replaced by $\tilde{a}_t$ to obtain the reference joint position.

After the PD torque is computed, only the torque of the selected failed joint is disabled,

$$\tau_t(j) = \begin{cases} 0, & \text{if joint } j \text{ is failed,} \\ \tau_t^{\text{PD}}(j), & \text{otherwise,} \end{cases} \quad (5)$$

where $\tau_t(j)$ denotes the applied torque after joint masking and $\tau_t^{\text{PD}}(j)$ is the torque generated by the PD controller. The failed joint therefore moves passively according to gravity, robot motion, and terrain contact.

The remaining functional joints in the affected leg retain actuation and move the leg toward the folded posture through the modified action. This keeps the affected leg closer to the body and reduces unintended contact with terrain edges while allowing the policy to coordinate the remaining functional joints for recovery. For a single-joint failure, the folded action is applied to one affected leg. For a diagonal two-joint failure, it is applied to both affected legs.

D. Reward-Aware Population Allocation

The failure and terrain conditions introduced in the previous section can differ substantially in learning difficulty. Under uniform population allocation, each category receives the same number of parallel environments regardless of its current performance. As a result, categories that remain difficult may receive insufficient training experience, while

¹The predefined folded actions are 0, 0.7708, and -1.118 for the hip roll, hip pitch, and knee pitch joints, respectively.

Algorithm 1: Reward-Aware Population Allocation

Input: Number of categories K , number of parallel environments N ($N \geq K$), moving average returns $\bar{\mathbf{R}}$, category factors $\mathbf{g}$, scaling factors α , and small constant ε

Output: Population vector $\mathbf{n} \in \mathbb{N}^K$

for $i \leftarrow 1$ **to** K **do**

$$d_i \leftarrow \frac{g_i}{\alpha_i \bar{R}_i + \varepsilon};$$

end

Normalize $\mathbf{d}$ to obtain population ratios $\mathbf{p}$;

for $i \leftarrow 1$ **to** K **do**

$$n_i \leftarrow 1 + \lfloor (N - K)p_i \rfloor;$$

end

$$r \leftarrow N - \sum_{i=1}^K n_i;$$

Assign one additional environment to the r categories with the largest fractional remainders of $(N - K)p_i$;

return $\mathbf{n}$;

well-learned categories continue to occupy the same fraction of the training population.

We introduce Reward-Aware Population Allocation (RPA) to adjust the training population according to the current performance of each category. Failure and terrain populations are allocated independently using the same procedure. Each failure configuration and each terrain configuration defined in Sec. III.C is treated as an independent allocation category.

For either allocation, let K denote the number of categories and $i \in \{1, \dots, K\}$ the category index. Let $\bar{R}_i$ denote the moving average return of category i . The priority score is defined as

$$d_i = \frac{g_i}{\alpha_i \bar{R}_i + \varepsilon}, \quad (6)$$

where α_i is a positive scaling factor and ε is a small positive constant. A category with a lower return therefore receives a larger priority score.

For failure allocation, the category factor is set to one. For terrain allocation, the current terrain progress is additionally considered,

$$g_i = \begin{cases} 1, & \text{for failure allocation,} \\ (L_i + 1)^{-1}, & \text{for terrain allocation,} \end{cases} \quad (7)$$

where L_i denotes the current progress level of terrain category i . Thus, terrain categories with both lower returns and lower progress receive higher priority.

The priority scores are normalized to obtain the target population ratio,

$$p_i = \frac{d_i}{\sum_{k=1}^K d_k}, \quad (8)$$

where p_i denotes the target fraction of parallel environments assigned to category i . Algorithm 1 summarizes how the target ratios are converted into integer population sizes.

Training begins with uniform population allocation so that the policy can acquire basic locomotion behaviors and obtain

initial return estimates. RPA is then applied at fixed intervals to update the failure and terrain populations according to their current performance.

Because the failure and terrain populations are computed independently, RPA does not require separate performance statistics for every failure–terrain pair. At each environment reset, one failure configuration and one terrain configuration are assigned according to their respective population distributions. As training progresses, these distributions shift toward categories that remain difficult for the current policy, providing them with additional training experience.

E. Failure-Aware Policy Conditioning

A unified policy must adapt its behavior to the current failure condition without switching between separate controllers. This requires the policy to infer the current joint state from onboard observations.

A single observation may not provide sufficient information for this inference. For example, a large joint tracking error can result from terrain contact, an external disturbance, or joint torque loss, producing similar instantaneous observations. In contrast, a joint failure produces persistent changes in robot motion over multiple time steps. We therefore use the observation history o_t^H to infer both the current joint failure state and body motion.

The history encoder is based on a β -VAE latent-variable model. It maps the observation history o_t^H to a latent representation z_t that captures temporal information about the recent robot state. The latent representation is then used by a body-velocity estimator and a fault-state classifier. KL regularization encourages a compact latent distribution.

The history encoder and prediction heads are trained using four objectives: body-velocity estimation, latent regularization, observation reconstruction, and joint fault classification. The joint fault classification loss is defined as

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{12} \sum_{j=1}^{12} [y_{t,j} \log \hat{y}_{t,j} + (1 - y_{t,j}) \log(1 - \hat{y}_{t,j})], \quad (9)$$

where $y_{t,j} \in \{0, 1\}$ is the ground-truth fault label of joint j at time step t , and $\hat{y}_{t,j}$ is its predicted fault probability. A value of one indicates a failed joint, whereas zero indicates a functional joint.

The total training loss is

$$\begin{aligned} \mathcal{L}_{\text{cls}} = & \text{MSE}(\hat{v}_t, v_t) + \beta D_{\text{KL}}(q(z_t | o_t^H) \| p(z_t)) \\ & + \text{MSE}(\hat{o}_t, o_t) + \mathcal{L}_{\text{BCE}}, \end{aligned} \quad (10)$$

where v_t and $\hat{v}_t$ denote the ground-truth and estimated body velocities, respectively, and o_t and $\hat{o}_t$ denote the current and reconstructed observations. The distribution $q(z_t | o_t^H)$ is the latent distribution inferred from the observation history, and $p(z_t)$ is its prior distribution. The term D_{KL} denotes the Kullback–Leibler divergence, β controls the strength of latent regularization, and MSE denotes mean squared error. The ground-truth body velocity and joint fault labels are available

only during simulation training and are not required during deployment.

The fault-state classifier produces a fault probability for each of the twelve actuated joints, forming the vector $\hat{\mathbf{y}}_t$. The estimated body velocity $\hat{v}_t$ and the predicted fault-state vector $\hat{\mathbf{y}}_t$ are provided to the actor together with the current policy observation. The unified policy is therefore written as

$$a_t = \pi_\theta(o_t, \hat{v}_t, \hat{\mathbf{y}}_t), \quad (11)$$

where $\hat{v}_t$ provides history-based body-motion information and $\hat{\mathbf{y}}_t$ represents the inferred state of the actuated joints.

The same actor is used during normal operation, single-joint failures, and selected diagonal two-joint failures. When a failure occurs, its effect appears in the observation history, causing the estimated body velocity and fault-state probabilities to change accordingly. The actor can therefore adapt its locomotion behavior using the updated estimates without explicit switching between failure-specific policies.

IV. EXPERIMENTAL RESULTS

We evaluate the proposed framework through simulation and real-robot experiments. The experiments focus on three aspects. First, we evaluate whether a single policy can maintain locomotion under normal, single-joint, and selected diagonal two-joint failure conditions. Second, we examine whether Reward-Aware Population Allocation reduces performance differences across failure and terrain categories. Third, we evaluate whether the learned recovery behaviors transfer to a real robot without additional real-robot training.

A. Simulation Setup

All policies were trained and evaluated in Isaac Gym [24]. The simulated platform was a Unitree Go2 robot with twelve actuated joints, with three actuated joints per leg.

Training configuration. The training and evaluation environments included six terrain configurations: *Flat*, *Discrete*, *Smooth Slope*, *Rough Slope*, *Stairs Up*, and *Stairs Down*. The failure conditions included twelve single-joint torque-loss cases and selected diagonal two-joint torque-loss cases. For each failure condition, Random Joint Masking was applied to the selected failed joint or joints. The folded-leg posture described in Sec. III.C was applied to each affected leg to reduce unwanted contact with stair edges and other obstacles.

Training began with uniform population allocation. After the proposed policy acquired basic locomotion behaviors, Reward-Aware Population Allocation was used to update the failure and terrain populations. Domain randomization and sensor noise remained active throughout training.

Comparison and evaluation. We compared the proposed method with BASELINE [22], FT-NET [9], FMT [23], and MASKING [12]. For evaluation, each terrain and failure combination was tested using 100 parallel environments for 1000 simulation steps. The primary metric was the root-mean-square error (RMSE) between the commanded and measured body velocities.

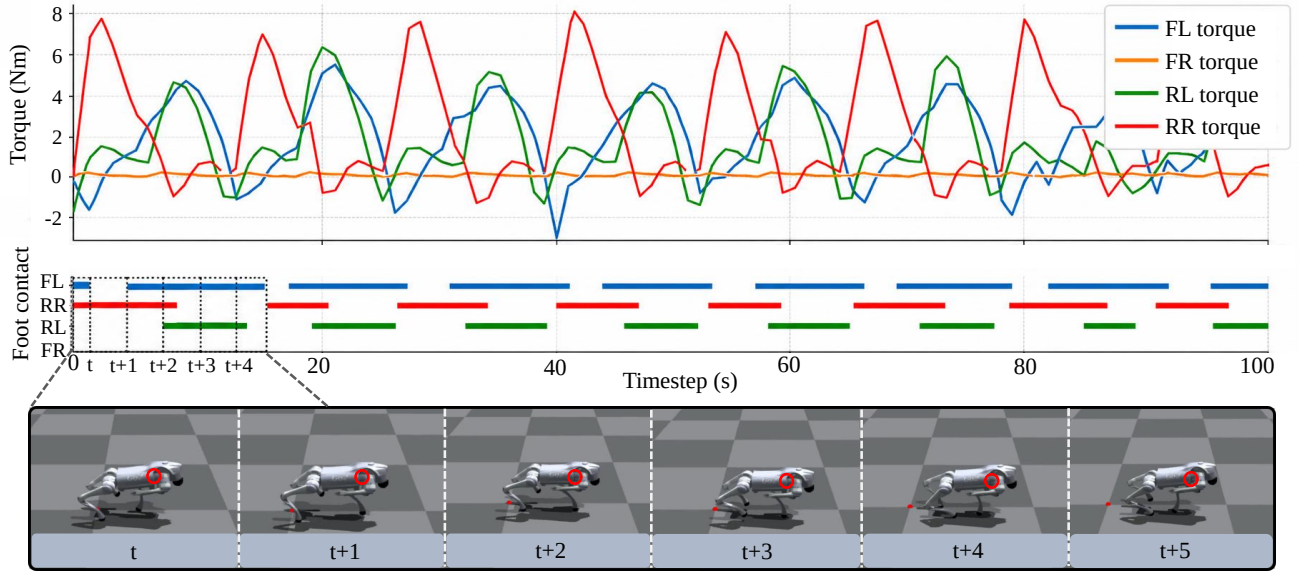


Fig. 4: Mean leg torque and foot-contact patterns under a single-joint torque-loss condition. Each torque curve represents the mean torque across the three joints of one leg. The functional legs adjust their torque and contact timing to maintain forward locomotion. The mean leg torques remain below 8 Nm without repeated saturation, while the foot-contact pattern remains repeatable across steps but differs from the nominal trot.

TABLE II: Tracking RMSE at a command velocity of 1 m/s across different terrain conditions. For each terrain, the results are averaged over the evaluated failure states within each failure group. Lower values indicate better tracking performance. The best and second-best results among the compared methods are shown in **bold** and underlined, respectively. A dash indicates that the corresponding method was not evaluated under the selected diagonal two-joint torque-loss conditions.

Failure condition	Terrain	RMSE (Command velocity = 1 m/s)				
		BASELINE [22]	FT-NET [9]	FMT [23]	MASKING [12]	Ours
Normal	Smooth Slope	0.1432	0.3039	0.2242	0.2207	<u>0.2192</u>
	Rough Slope	0.1681	0.3183	0.3742	0.2606	<u>0.2430</u>
	Stairs Up	<u>0.8430</u>	0.8618	0.8910	0.8989	0.2269
	Stairs Down	<u>0.4163</u>	0.8632	0.6423	0.7921	0.2340
	Discrete	0.7815	0.6514	<u>0.6193</u>	0.6234	0.2257
	Avg	<u>0.4704</u>	0.5997	<u>0.5502</u>	0.5591	0.2298
Single-joint torque loss	Smooth Slope	0.9339	<u>0.2636</u>	0.3015	0.2722	0.2408
	Rough Slope	0.9473	0.3608	0.4300	<u>0.2830</u>	0.2561
	Stairs Up	0.9858	<u>0.8266</u>	0.8883	0.8992	0.2904
	Stairs Down	0.9482	<u>0.5074</u>	0.7527	0.6573	0.2638
	Discrete	0.9605	<u>0.6329</u>	0.7015	0.6640	0.2569
	Avg	0.9551	<u>0.5182</u>	0.6148	0.5551	0.2616
Diagonal two-joint torque loss	Smooth Slope	—	—	—	—	0.2901
	Rough Slope	—	—	—	—	0.3138
	Stairs Up	—	—	—	—	0.3237
	Stairs Down	—	—	—	—	0.3213
	Discrete	—	—	—	—	0.3056
	Avg	—	—	—	—	0.3109

B. Performance Across Failure and Terrain Conditions

Table II compares tracking performance across failure and terrain conditions. The proposed policy maintains relatively consistent tracking performance as the terrain and failure conditions change.

Performance across terrains. Under normal operation, BASELINE achieves lower RMSE than the proposed method on *Smooth Slope* and *Rough Slope*. This suggests that a policy optimized primarily for nominal locomotion can

achieve lower tracking error under regular contact conditions. However, its performance degrades on terrains with abrupt height changes. For example, the RMSE of BASELINE increases to 0.8430 on *Stairs Up*, whereas the proposed policy achieves 0.2269.

Similar performance degradation is observed for the other comparison methods on *Stairs Down* and *Discrete* terrain. In contrast, the proposed policy maintains similar RMSE values across the evaluated terrain categories. Its main benefit is therefore the reduced performance degradation on discontin-

TABLE III: Tracking RMSE on flat ground under normal operation and single-joint torque-loss conditions. Lower values indicate better tracking performance. The best and second-best results in each row are shown in **bold** and underlined, respectively.

Failure condition	Failure location		RMSE ↓				
			BASELINE [22]	FT-NET [9]	FMT [23]	MASKING [12]	Ours
Normal	None		0.1469	<u>0.1142</u>	0.2066	0.1699	0.1032
Single-joint torque loss	Front left	HR	0.7089	0.2739	0.3162	<u>0.1507</u>	0.1048
		HP	0.9747	0.2771	0.2842	<u>0.2668</u>	0.1081
		KP	0.9252	0.2927	0.2751	<u>0.1920</u>	0.1059
	Avg		0.8696	0.2812	0.2918	<u>0.2031</u>	0.1063
	Rear right	HR	0.8928	0.2652	0.2753	<u>0.1986</u>	0.1018
		HP	1.0320	0.2741	0.2733	<u>0.1959</u>	0.1023
		KP	0.8081	0.2499	0.2121	<u>0.1492</u>	0.1101
	Avg		0.9109	0.2631	0.2536	<u>0.1812</u>	0.1047

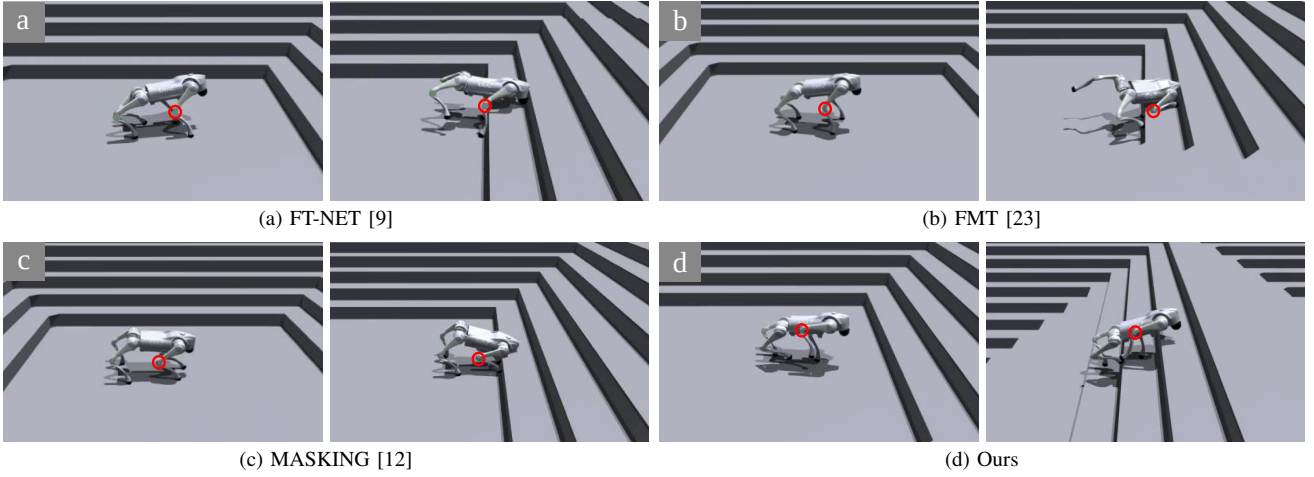


Fig. 5: Stair ascent under the same front-right knee-pitch torque-loss condition. Each panel shows an early and a later stage of the ascent. In (a)–(c), FT-NET, FMT, and MASKING lose balance while traversing the stair edges. In (d), the proposed policy completes the ascent while keeping the affected leg close to the body.

uous terrain rather than the lowest tracking error on every continuous terrain.

The difference becomes more apparent after joint torque loss. The average RMSE of BASELINE increases from 0.4704 under normal operation to 0.9551 under single-joint torque loss. In contrast, the average RMSE of the proposed method changes from 0.2298 to 0.2616. These results indicate a smaller performance degradation after joint failure. The same actor is used across normal and failed conditions without switching to a failure-specific controller.

Performance across joint failures. Table III reports detailed results for representative front-left and rear-right single-joint failures on flat ground. The proposed policy achieves RMSE values between 0.1018 and 0.1101 across hip-roll, hip-pitch, and knee-pitch failures. The narrow RMSE range indicates relatively consistent tracking performance across the evaluated joint locations and joint types. In contrast, the comparison methods show larger performance variations across joint conditions.

The selected diagonal two-joint failures are more challenging than the single-joint failures. Their average RMSE across the evaluated terrains is 0.3109, compared with 0.2616 under single-joint failures. The increase in RMSE reflects

the greater difficulty of coordinating locomotion under simultaneous failures on opposing diagonal legs. Nevertheless, the policy maintains locomotion across all evaluated terrain categories.

Recovery behavior. Fig. 4 shows a representative recovery behavior under a single-joint torque-loss condition. Each torque curve represents the mean torque of the three joints in one leg. The functional legs adjust their torque and contact timing to maintain forward locomotion, while the mean torque remains within the shown range without repeated saturation.

The foot-contact pattern remains repeatable across several steps but differs from the nominal trot. The policy adjusts contact timing and support duration according to the failure condition, resulting in a recovery pattern that differs from the nominal gait.

Fig. 5 compares stair ascent under the same front-right knee-pitch torque-loss condition. FT-NET, FMT, and MASKING lose balance while traversing the stair discontinuities, whereas the proposed policy completes the ascent while keeping the affected leg close to the body.

This comparison illustrates two forms of adaptation during locomotion under joint failure on discontinuous terrain. The

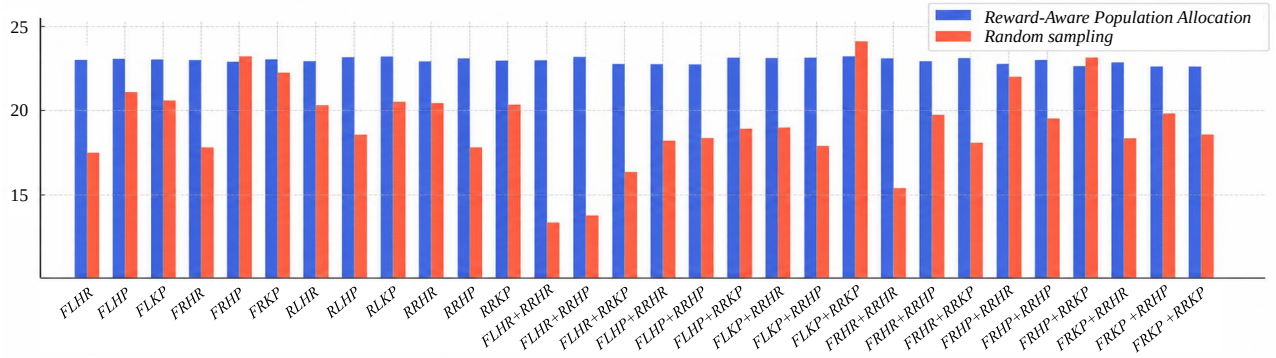


Fig. 6: Comparison of uniform allocation and failure population allocation. The bars show the final episodic return of each failure category. Reward Aware Population Allocation increases the returns of weak failure categories and reduces the performance difference across failure conditions.

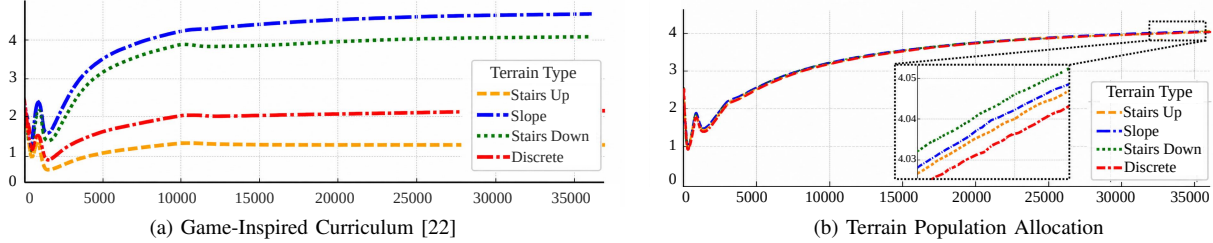


Fig. 7: Learning curves for representative terrain categories. The plots show the episodic return during training on *Stairs Up*, *Slope*, *Stairs Down*, and *Discrete* terrain. In (a), the baseline curriculum results in substantial differences in return across terrain categories. In (b), terrain population allocation reduces these differences, leading to a narrower return range across the evaluated terrains.

functional legs adjust their motion to support the body, while the folded-leg posture reduces unwanted contact between the affected leg and stair edges.

C. Effect of Failure Population Allocation

The failure component of Reward-Aware Population Allocation assigns more parallel environments to failure categories with lower returns. Its objective is to reduce performance imbalance across failure conditions rather than simply maximize the average return.

Fig. 6 compares the final episodic return of each failure category under uniform allocation and failure population allocation. Under uniform allocation, the final returns range from 11.7 to 24.2, resulting in a return spread of 12.5. This large variation indicates that equal population allocation does not lead to uniform learning progress across failure categories.

With Reward-Aware Population Allocation, the final returns converge to a much narrower range of 22.8–23.3, reducing the return spread to 0.5. The mean return also increases from 18.96 to 23.05. The largest improvements occur in categories that exhibit the lowest returns under uniform allocation.

These results indicate that failure categories differ substantially in learning difficulty even when they receive the same number of parallel environments. Reward-Aware Population Allocation shifts training experience toward lower-performing categories, reducing the performance gap across failure conditions. This redistribution prioritizes balanced performance across failure categories rather than further improving conditions that are already well learned.

TABLE IV: Fault-state classification performance using MSE and BCE training objectives. Higher values indicate better classification performance.

Loss Type	Accuracy	Precision	Recall	F1 Score
MSE loss	0.8203	0.8333	0.8201	0.8291
BCE loss	0.9820	0.9780	0.9800	0.9790

D. Effect of Terrain Population Allocation

Terrain categories introduce another source of training imbalance because different terrains exhibit different learning rates. Fig. 7(a) shows the learning curves under the baseline terrain curriculum. The returns for *Slope* and *Stairs Down* increase earlier and reach higher values, whereas *Stairs Up* and *Discrete* remain at lower returns. This difference indicates that a fixed terrain allocation can lead to uneven learning progress across terrain categories.

Fig. 7(b) shows the result of terrain population allocation. After the initial uniform allocation stage, more environments are assigned to terrain categories with lower returns and lower terrain progress. As training proceeds, the learning curves move closer together and converge within a narrower return range. This behavior indicates that terrain population allocation redistributes training experience toward terrain categories that progress more slowly.

The resulting reduction in performance differences is consistent with the tracking results in Table II. The proposed policy does not achieve the lowest RMSE on every smooth terrain under normal operation. However, its performance degrades less on *Stairs Up*, *Stairs Down*, and *Discrete* terrain. These results suggest that terrain population allocation promotes more balanced learning across terrain categories rather than concentrating training on conditions that are learned more quickly.

TABLE V: Ablation results at a command velocity of 1 m/s. A, B, and R indicate Asymmetric Actor-Critic (AAC), Binary Cross-Entropy (BCE), and Reward-Aware Population Allocation (RPA), respectively. FL and RR denote the front-left and rear-right single-joint failure groups, while FL/RR and FR/RL denote the selected diagonal two-joint failure groups. The values report tracking RMSE, where lower is better. The best result in each column is shown in **bold**.

Training configuration	Components			Tracking RMSE ↓					
	A	B	R	Normal	FL	RR	FL/RR	FR/RL	Overall Avg
MASKING	×	×	×	0.1699	0.2031	0.1812	0.4576	0.2868	0.2831
MASKING+AAC	✓	×	×	0.1208	0.1758	0.1670	0.4889	0.4421	0.3432
MASKING+AAC+BCE	✓	✓	×	0.1050	0.1600	0.1627	0.4176	0.4267	0.3179
MASKING+BCE	×	✓	×	0.1730	0.2232	0.1689	0.5118	0.4743	0.3704
MASKING+RPA	×	×	✓	0.2231	0.2294	0.2348	0.2841	0.2841	0.2633
MASKING+BCE+RPA	×	✓	✓	0.1801	0.1946	0.1855	0.2427	0.2518	0.2253
MASKING+AAC+RPA	✓	×	✓	0.1201	0.1337	0.1441	0.1830	0.1825	0.1652
Ours	✓	✓	✓	0.1032	0.1057	0.1047	0.1313	0.1450	0.1273

E. Fault-State Classification

Table IV compares MSE and BCE training objectives for joint fault-state classification. The classifier trained with BCE improves accuracy from 0.8203 to 0.9820, while the F1 score increases from 0.8291 to 0.9790. Precision and recall also increase to 0.9780 and 0.9800, respectively.

A classifier trained with MSE produces continuous outputs and may yield values near the decision boundary between functional and failed states. Such ambiguous predictions make the fault-state input provided to the actor less distinct. In contrast, training with BCE directly encourages the classifier to distinguish functional and failed states for each joint, resulting in clearer separation between the two classes.

The higher recall reduces false negatives in joint-failure detection, whereas the higher precision reduces false positives for functional joints. Table IV evaluates classification performance only and does not independently quantify its effect on locomotion performance. The contribution of fault-state classification to control performance is examined separately in the ablation study.

F. Ablation Study

Table V evaluates the contributions of Asymmetric Actor-Critic (AAC), the BCE training objective for fault-state classification, and Reward-Aware Population Allocation (RPA). These components address different aspects of the learning framework.

Applying RPA alone reduces the overall RMSE from 0.2831 to 0.2633. Although the RMSE under normal and single-joint conditions increases slightly, larger improvements are observed under the selected diagonal two-joint failures. For example, the average RMSE of the first evaluated diagonal failure group decreases from 0.4576 to 0.2841. This result is consistent with the objective of RPA, which redistributes training experience toward lower-performing failure categories rather than improving all conditions uniformly.

Combining AAC with RPA further reduces the overall RMSE to 0.1652 and improves performance across nor-

mal, single-joint, and diagonal two-joint conditions. RPA increases the training exposure of difficult conditions, while the asymmetric critic uses privileged simulation information during training. The improvement obtained by their combination indicates that population reallocation and privileged critic information provide complementary benefits during policy optimization.

Training the fault-state classifier with BCE alone does not reduce the overall tracking error. However, when BCE is added to AAC and RPA, the overall RMSE decreases further from 0.1652 to 0.1273. As shown in Table IV, training with BCE also produces more accurate fault-state predictions than training with MSE. These results indicate that more accurate fault-state information becomes particularly useful when combined with the other components of the framework.

The complete framework achieves an overall RMSE of 0.1273, approximately 55% lower than that of MASKING. Overall, RPA reduces performance imbalance across failure categories, AAC provides privileged information to the critic during training, and BCE improves the accuracy of fault-state classification. Their combination yields the lowest overall tracking error across the evaluated failure conditions.

G. Real-Robot Validation

The real-robot experiments evaluate whether the recovery behaviors learned in simulation transfer to contact conditions that differ from the simulated training environments. The policy was deployed on a Unitree Go2 robot without additional real-robot training. We evaluated representative single-joint and selected diagonal two-joint torque-loss conditions on a deformable rubber surface, uneven wooden planks, a dynamic seesaw, and outdoor stairs. The experiments are summarized in Fig. 8.

Deformable and uneven terrain. Fig. 8(a)–(d) show locomotion on the deformable rubber surface and uneven wooden planks. The rubber surface introduces compliance and changes the local contact response, while the wooden planks create uneven foothold heights and abrupt contact

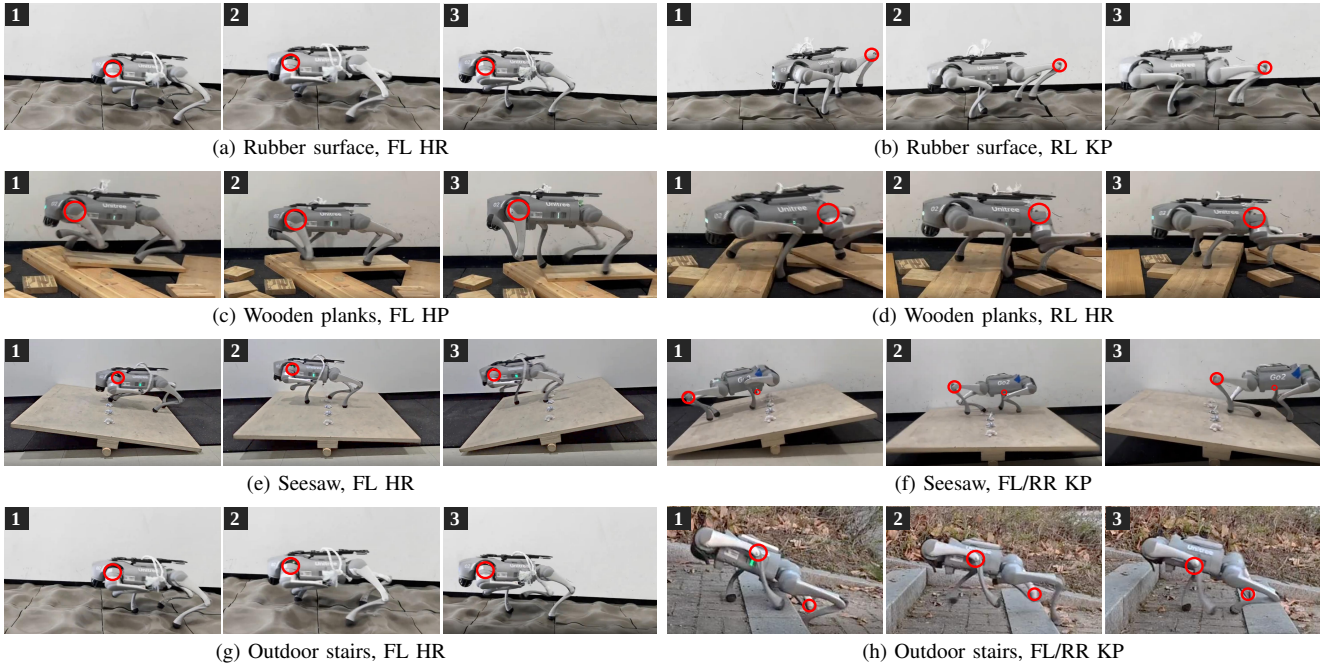


Fig. 8: Real-robot validation across deformable, uneven, moving, and discontinuous terrains. The Unitree Go2 traverses a deformable rubber surface in (a) and (b), uneven wooden planks in (c) and (d), a dynamic seesaw in (e) and (f), and outdoor stairs in (g) and (h). The experiments include representative single-joint and selected diagonal two-joint torque-loss conditions. FL, RL, and RR denote the front-left, rear-left, and rear-right legs, respectively, while HR, HP, and KP denote the hip-roll, hip-pitch, and knee-pitch joints.

changes. On the rubber surface, the robot maintains locomotion under front-left hip-roll and rear-left knee-pitch failures. On the wooden planks, it maintains locomotion under front-left hip-pitch and rear-left hip-roll failures. In each case, the affected leg remains close to the body while the functional legs continue to support locomotion over the varying contact conditions. These experiments show that the learned recovery behavior transfers to surface properties and foothold variations that differ from those used during training.

Dynamic seesaw. Fig. 8(e) and (f) show locomotion on a dynamic seesaw. The platform changes its direction of tilt as the robot crosses the central pivot, resulting in rapid changes in platform orientation, body attitude, and contact timing. The single-joint experiment uses a front-left hip-roll failure, whereas the diagonal two-joint experiment uses front-left and rear-right knee-pitch failures. In both cases, the robot crosses the central pivot and completes the traversal despite the motion of the support surface. Unlike the static terrain profiles used during training, the seesaw changes its orientation in response to contact. The results therefore provide qualitative evidence that the learned recovery behavior transfers to dynamically changing support conditions.

Outdoor stairs. Fig. 8(g) and (h) show the outdoor stair experiments. Repeated height changes increase the likelihood of contact between an affected leg and stair edges, making this terrain particularly challenging under joint failure. In the single-joint condition, the front-left hip-roll joint is disabled, and the robot completes the ascent without falling. In the diagonal two-joint condition, the front-left and rear-right knee-pitch joints are disabled. The robot continues to place

the functional legs on successive steps and completes the stair sequence.

During ascent, the folded-leg posture keeps the affected leg close to the body and reduces unwanted interaction with stair edges. However, successful stair traversal also requires the policy to control the body and coordinate the remaining functional legs under changing support conditions. The experiment therefore evaluates the combined effect of the folded-leg posture and the learned recovery behavior on discontinuous terrain.

Overall, the real-robot experiments show that the proposed policy maintains locomotion on deformable, uneven, moving, and discontinuous terrain without additional real-robot training. The hardware evaluation is qualitative and does not include repeated success rates or statistical comparisons with the other methods. The results should therefore be interpreted as evidence of practical sim-to-real transfer rather than a complete measure of real-robot reliability.

V. CONCLUSION

In this paper, we presented a unified learning framework for fault-tolerant quadrupedal locomotion across diverse joint-failure and terrain conditions. The framework enables a single policy to handle normal operation, single-joint failures, and selected diagonal two-joint failures across multiple terrain types. Simulation and real-robot experiments showed that the learned policy maintains locomotion under these failure conditions on continuous and discontinuous terrain, including stairs, without additional real-robot training.

The current study considers only a selected set of diagonal two-joint failure conditions and therefore does not cover

the full space of multiple-joint failures. Some simultaneous failure configurations, such as failures affecting both front or both rear legs, may require failure-specific initialization or an additional transient recovery phase before stable locomotion can be established. Extending the framework to a broader range of simultaneous failures will require further investigation of recovery initialization and policy scalability, including how network capacity should scale as the number and diversity of failure conditions increase. Future work will also include repeated hardware evaluations and partial actuator degradation.

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